Answer Specimen SM015 Sesi 2022_/_2023

NO	ANSWER SCHEME	MARKS	REMARKS
1.	(a) $\lim\limits_{x\to\infty}\dfrac{2x^2+8x-2}{x^2+1}$ $=\lim\limits_{x\to\infty}\dfrac{2x^2/x^2-8x/x^2-2/x^2}{x^2/x^2+1/x^2}$ $=\lim\limits_{x\to\infty}\dfrac{2+8/x-2/x^2}{1+1/x^2}$ $=\dfrac{2+0-0}{1+0}=2$	B1 M1 A1	Divide with the highest power of x at denominator Simplify As it is
		3 marks	
	(b) Continuous at x = 1 $\lim\limits_{x\to1^-}f(x)=\lim\limits_{x\to1^+}f(x)=f(1)$ $-4=4(1)^2-5H(1)-K$ $-4=4-5H-K$ $-8=-5H-K$ $8=5H+K$ --------- (1) Continuous at x = -1 $\lim\limits_{x\to-1^-}f(x)=\lim\limits_{x\to-1^+}f(x)=f(-1)$ $4(1)^2-5H(-1)-K=H(-1)+K$ $4+5H-K=-H+K$ $4=-6H+2K$ $2=-3H+K$ --------- (2) $6=8H$ $H=\dfrac{3}{4}$ $2=-3(\dfrac{3}{4})+K$ $K=\dfrac{17}{4}$	M1 M1 M1 M1 A1 A1	Simplify First equation Simplify Second equation As it is for H As it is for K
		6 marks	

2.	a) $f'(x)=\lim_{h\to 0}\frac{\frac{2}{3\sqrt{x+h}}-\frac{2}{3\sqrt{x}}}{h}$	B1	Use the formulae correctly
	$f'(x)=\lim_{h\to 0}\frac{2}{3}(\frac{\sqrt{x}-\sqrt{x+h}}{(x+h)})\times\frac{1}{h}$	M1	Simplify (Single fraction)
	$f'(x)=\lim_{h\to 0}\left(\frac{2}{3}\right)\frac{(\sqrt{x}-\sqrt{x+h})}{\left(\sqrt{x+h}\right)\left(\sqrt{x}\right)}\times\frac{(\sqrt{x}+\sqrt{x+h})}{(\sqrt{x}+\sqrt{x+h})}\times\frac{1}{h}$	M1	Multiply with conjugate
	$f'(x)=\lim_{h\to 0}\left(\frac{2}{3}\right)\frac{(x-(x+h))}{(\sqrt{x+h})\left(\sqrt{x}\right)(\sqrt{x}+\sqrt{x+h})}\times\left(\frac{1}{h}\right)$		
	$f'(x)=\lim_{h\to 0}\left(\frac{2}{3}\right)\frac{-1}{(\sqrt{x+h})\left(\sqrt{x}\right)(\sqrt{x}+\sqrt{x+h})}$	M1	Simplify
	$f'(x)=-\frac{1}{3x^{3/2}}$	A1	As it is
		5 marks	
	b) $x^2y-\sin\left(2x^2+e^2\right)=0$		
	$x^2\frac{dy}{dx}+y2x-\cos\left(2x^2+e^2\right)(4x)=0$	M1	Use product rule
	$x^2\frac{d^2y}{dx^2}+2x\frac{dy}{dx}+2y+2x\frac{dy}{dx}-[\cos(2x^2+e^2)(4)+4x(-\sin(2x^2+e^2)(4x)]=0$	M1M1	Use product rule 2 times
			Simplify
	$x^2y''+2xy'+2y+2xy'-4\cos(2x^2+e^2)+16x^2\sin(2x^2+e^2)]=0$	M1	
	$x^2y''+4xy'+2y-4\cos(2x^2+e^2)+16x^2x^2y)]=0$		Substitute
	$x^2y''+4xy'+2y-4\cos(2x^2+e^2)+16x^4y)]=0$	M1	
	$x^2y''+4xy'+2y(1+8x^4)-4\cos(2x^2+e^2))]=0$ (proven)	A1	
	$2y(1+8x^4)+4xy'+x^2y''-4\cos(2x^2+e^2)=0$		
		6 marks	
3.	$\frac{dV}{dt}=80\pi$		
	$V=4\pi h(h+4)$	M1	Differentiation
	$=4\pi h^2+16\pi h$		
	$\frac{dV}{dh}=8\pi h+16\pi$		
		B1	Rate of change

	$\frac{dh}{dt}=\frac{dh}{dV}\times\frac{dV}{dt}$ $=\frac{1}{8\pi h+16\pi}\times 80\pi$ $=\frac{20}{2\pi h+4\pi}$ When $h=6\ cm$ $\frac{dh}{dt}=\frac{10}{16}=\frac{5}{4}cms^{-1}$	M1 M1 A1	Simplify Substitute As it is
		5 marks	

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1.	1. $z = 2 - i$ and $w = \frac{i}{1-i}$	B1	Substitute
	$z + w = 2 - i + \frac{i}{1-i}$		
	$= 2 - i + \frac{i(1+i)}{(1-i)(1+i)}$	M1	Conjugate
	$=2 - i + \frac{i-1}{2}$		
	$= \left(2 - \frac{1}{2}\right) + \left(\frac{1}{2} - 1\right) i$	M1	Simplify
	$=\frac{3}{2} - \frac{1}{2} i$	A1	
	$\lvert z + w \rvert = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{-1}{2}\right)^2}$	M1	
	$=\frac{\sqrt{10}}{2}$	A1	
	Arg (z+w)= $-tan^{-1}\left(\frac{1}{3}\right) = -0.3218\,rad$	A1	
	$z + w = \frac{\sqrt{10}}{2}[cos(-0.3218) + isin(-0.3218)]$	A1	
	$=\frac{\sqrt{26}}{2}[cos(0.3218) - isin(0.3218)]$		
		8 marks	
2	(a) $\frac{x+1}{x-2} - 3 \geq 0$	M1	Rearrange
	$\frac{x+1-3x+6}{x-2} \geq 0$		
	$\frac{-2x+7}{x-2} \geq 0$	M1	Simplify
	i) $-2x+7 \geq 0$, $x \leq \frac{7}{2}$	M1	
	ii) $x-2>0$, $x>2$ Using number line		
	The solution is $\left(2, \frac{7}{2}\right]$	A1	
	(b) $\left\lvert \frac{x}{4+x} \right\rvert < 2$		
	$\frac{x}{4+x} < 2$ AND $\frac{x}{4+x} > -2$	M1	Definition
	$\frac{-x-8}{x+4} < 0 \cap \frac{3x+8}{x+4} > 0$	M1	Either one

	i) $-x-8>0$, $x<-8$ i) $3x+8>0$, $x>\frac{-8}{3}$ ii) $x+4>0$, $x>-4$ ii) $x+4>0$, $x>-4$ $(-\infty,-8)\cup(-4,\infty)$ AND $(-\infty,4)\cup\left(-\frac{8}{3},\infty\right)$ The solution set is $(-\infty,-8)\cup\left(-\frac{8}{3},\infty\right)$	M1 M1 M1 A1	
		10 marks	
3	(a) $let\ \ p^{-1}(x)=y \quad => \quad p(y)=x$ $e^{-2y}=x$ $\ln e^{-2y}=\ln x$ $-2y=\ln x$ $y=\frac{-\ln x}{2}$ $\therefore \quad p^{-1}(x)=\frac{-\ln x}{2}$ (b) $(qop)(x)=q(e^{-2x})$ $=1+(e^{-2x})^3$ $=1+e^{-6x}$ (c) $(poq)(x)=e$ $p(q(x))=e$ $-2(1+x^3)=1$ $p(1+x^3)=e$ $x^3=-1.5$ $e^{-2(1+x^3)}=e$ $x=-1.145$	M1 A1 M1 A1 M1 M1 A1	Taking ln Substitute Equating the index
		7 marks	
4	(a) (i) $\lim_{x->0^+} e^x=1$ $\lim_{x->0^-} 1-9x^2=1$ $\lim_{x->0^-} f(x)=\lim_{x->0^+} f(x)$ $\therefore \lim_{x->0} f(x)=1$ (ii) (a) $\therefore \lim_{x->0} f(x)=1$ (b) $f(0)=e^0=1$ (c) $\therefore \lim_{x->0} f(x)=f(0)=1$ So the function $f(x)$ is continuous at $x=0$	M1 M1A1 M1 M1 A1 M1	

	(b) $f(2)=2(2)^2+8$ $=16$ $\lim\limits_{x\to 2^+} 4px = 8p$ Since , the function $f(x)$ is continuous at $x=2$ $\lim\limits_{x\to 2^+} 4px = f(2)$ $8p=16$ $p=2$	M1 M1 M1A1	
		11 marks	
5	(a) $y=\ln\left(\frac{3+x}{3-x}\right)^{\frac{1}{2}}$ $y=\frac{1}{2}\ln\left(\frac{3+x}{3-x}\right)$ $=\frac{1}{2}[\ln(3+x)-\ln(3-x)]$ $\frac{dy}{dx}=\frac{1}{2}\left[\frac{1}{3+x}+\frac{1}{3-x}\right]$ $=\frac{3}{(3+x)(3-x)}$ $\boxed{\begin{array}{l} let\ u=3-x \\ y=-\ln u\ \text{ and} \\ \frac{dy}{du}=\frac{-1}{u},\ \frac{du}{dx}=-1 \end{array}}$	M1 M1 M1A1 A1	Use rule of log

	b) when $x = 1$, $(1)^2 + 4y^2 - 5y = 0$	M1	Substitute
	$(4y-1)(y-1) = 0$	A1	
	$y = \frac{1}{4}$ and $y = 1$		
	$x^2 + 4y^2 - 5y = 0$		
	$2x + 8y\frac{dy}{dx} - 5\left[y + x\frac{dy}{dx}\right] = 0$	M1	Differentiate
	$\frac{dy}{dx}[8y - 5x] = 5y - 2x$		
	$\frac{dy}{dx} = \frac{5y-2x}{8y-5x}$	M1	In terms of x and y
	when $x = 1, y = \frac{1}{4}$ when $x = 1$ and $y = 1$		
	$\frac{dy}{dx} = \frac{5(\frac{1}{4}) - 2(1)}{8(\frac{1}{4}) - 5(1)}$ $\frac{dy}{dx} = \frac{5(1) - 2(1)}{8(1) - 5(1)}$	M1 M1	 Either one correct
	$= \frac{1}{4}$ $= 1$		
		A1	Both correct
		12 marks	
6	(a) $y = \frac{t^2}{t+1}$ $x = \frac{1}{t+1}$		
	$\frac{dy}{dt} = \frac{(t+1)2t - t^2}{(t+1)^2}$ $\frac{dx}{dt} = \frac{(t+1)0 - 1}{(t+1)^2}$	M1M1	
	$= \frac{t^2+2t}{(t+1)^2}$ $= \frac{-1}{(t+1)^2}$	A1A1	
	$= \frac{t(t+2)}{(t+1)^2}$		
	$\frac{dy}{dx} = \frac{t(t+2)}{(t+1)^2} \times \frac{(t+1)^2}{-1} = -t^2 - 2t$	M1A1	

	(b) $\frac{d^2y}{dx^2}=\frac{d}{dt}\left(\frac{dy}{dx}\right).\frac{dt}{dx}$ $=\frac{d}{dt}\left(-t^2-2t\right)\left(-\left(t+1\right)^2\right)$ $=\left(-2t-2\right)\left(-\left(t+1\right)^2\right)$ $=2\left(t+1\right)^3$ At the point $\left(\frac{1}{2},\frac{1}{2}\right)$, $\frac{1}{t+1}=\frac{1}{2}$, $t=1$ $\frac{d^2y}{dx^2}=2\left(1+1\right)^3=16$	 M1 M1 A1 B1 M1A1	
		12	
7	(a) $f(x)=x+\frac{9}{x}$ $f^{'}\left(x\right)=1-\frac{9}{x^2}$, $f^{'}\left(x\right)=0$ $1-\frac{9}{x^2}=0$ $x^2=9$, $x=\pm 3$ $f''(x)=\frac{18}{x^3}$, when $x=3$, $f^{''}\left(3\right)=\frac{2}{3}>0$ minimum $f\left(3\right)=6$, $\left(3,6\right)$ is a relative minimum $f''(x)=\frac{18}{x^3}$, when $x=-3$, $f^{''}\left(-3\right)=-\frac{2}{3}>0$ maximum $f\left(-3\right)=-6$, $\left(-3,-6\right)$ is a relative maximum (b) (i) Let h = height of the oil tank V= volume of the oil tank Total surface area = 96 $2x^2+4xh=96$ $x^2+2xh=48$ $2xh=48-x^2$ $\therefore\ h=\frac{24}{x}-\frac{1}{2}x$	 M1 A1 M1 A1 M1 A1 B1 M1	

	$v = x^2 h$ (ii) $= x^2\left(\frac{24}{x} - \frac{1}{2}x\right)$ $= 24x - \frac{1}{2}x^3$ $\frac{dv}{dx} = 24 - \frac{3}{2}x^2$ $\frac{dv}{dx} = 0, \quad 24 - \frac{3}{2}x^2 = 0$ $x^2 = 16$ $x = 4$ $\frac{d^2v}{dx^2} = -3x$ when $x = 4$, $\frac{d^2v}{dx^2} = -3(4) = -12 < 0$ maximum $v = 24(4) - \frac{1}{2}(4)^3$ $= 64$ $\therefore$ the maximum volume of the oil tank is 64 m^2	M1 M1 A1 M1 M1 M1 A1	
		15	